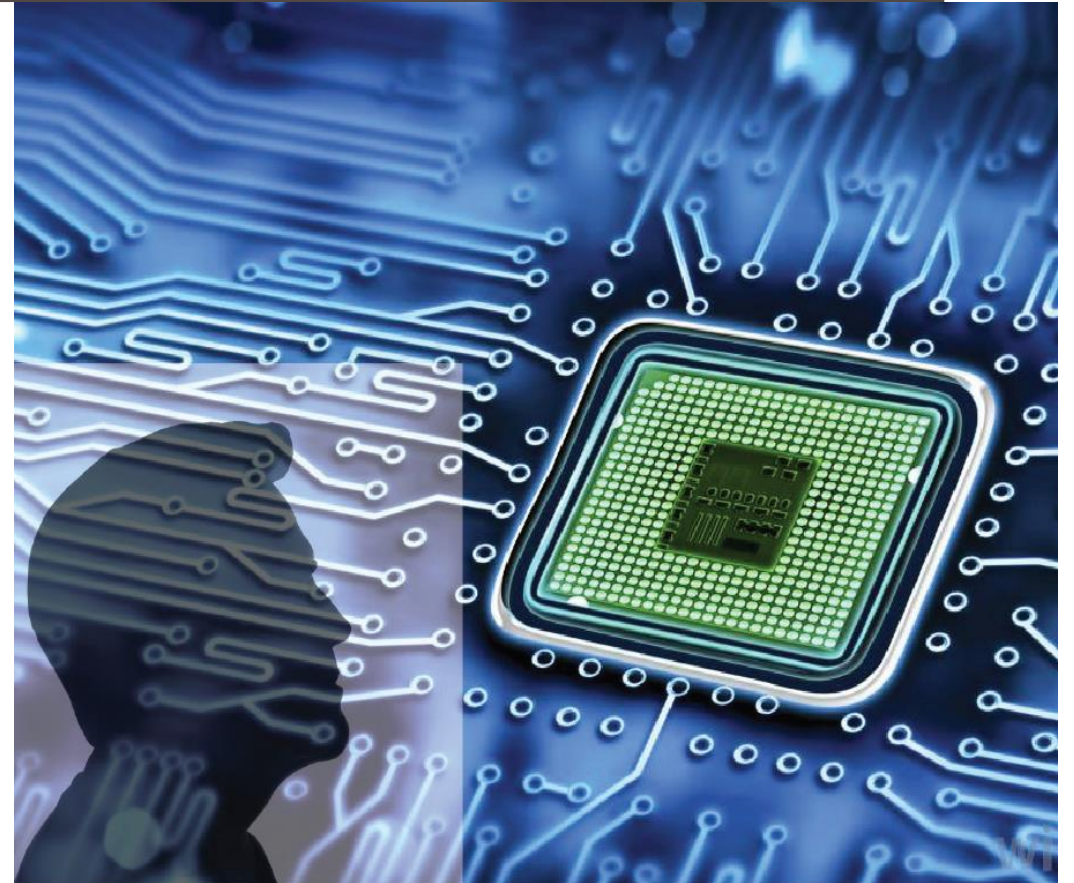


VLSI DESIGN FLOW

Week-4, Lecture-1 Introduction

Sneh Saurabh
2nd September, 2024



VDF: Revision/Announcements



Overview of RTL to GDS Flow



Verification

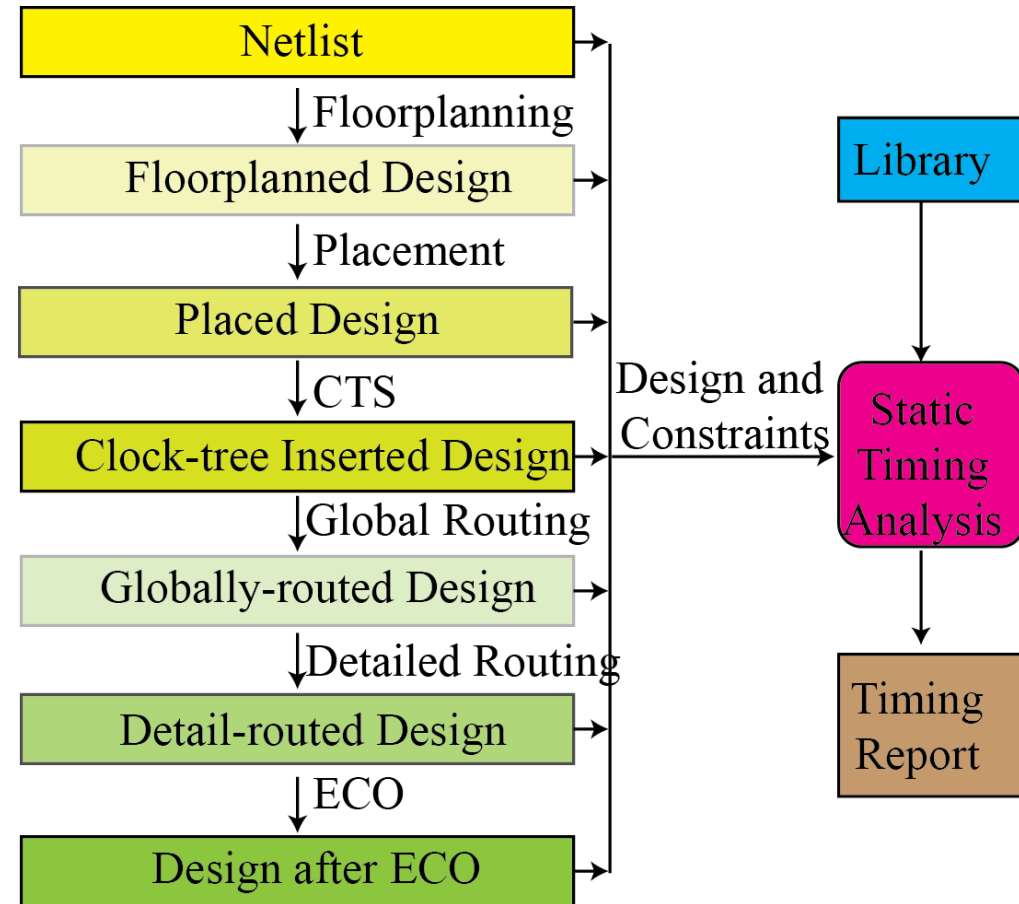
Verification Techniques: Static Timing Analysis

How to ensure the synchronicity of data transfer between flip-flops in a synchronous design?

Synchronicity: data launched by a flip-flop gets captured in the sequentially adjacent flip-flop in *the next clock cycle*

Static Timing Analysis (STA): ensures deterministic synchronous timing behavior in a circuit

- STA tools consider the worst-case behavior (may be pessimistic), but will always ensure the timing safety
- Carried out multiple times in a design flow



Verification Techniques: Physical Design

How to ensure that the layout does not have manufacturing and connectivity issues and the yield for a design remains high?

Physical verification:

- Check a set of rules during physical design before sending the layout to the foundry.

Design rule check (DRC):

- Rules are defined by the foundries and depend on the manufacturing technology
- All DRC violations must be fixed before sending it to the foundry for fabrication.

Electrical rule check (ERC):

- Rules are defined to ensure proper connectivity (for e.g.: no short circuit between distinct signal lines).

Layout vs. schematic (LVS) check:

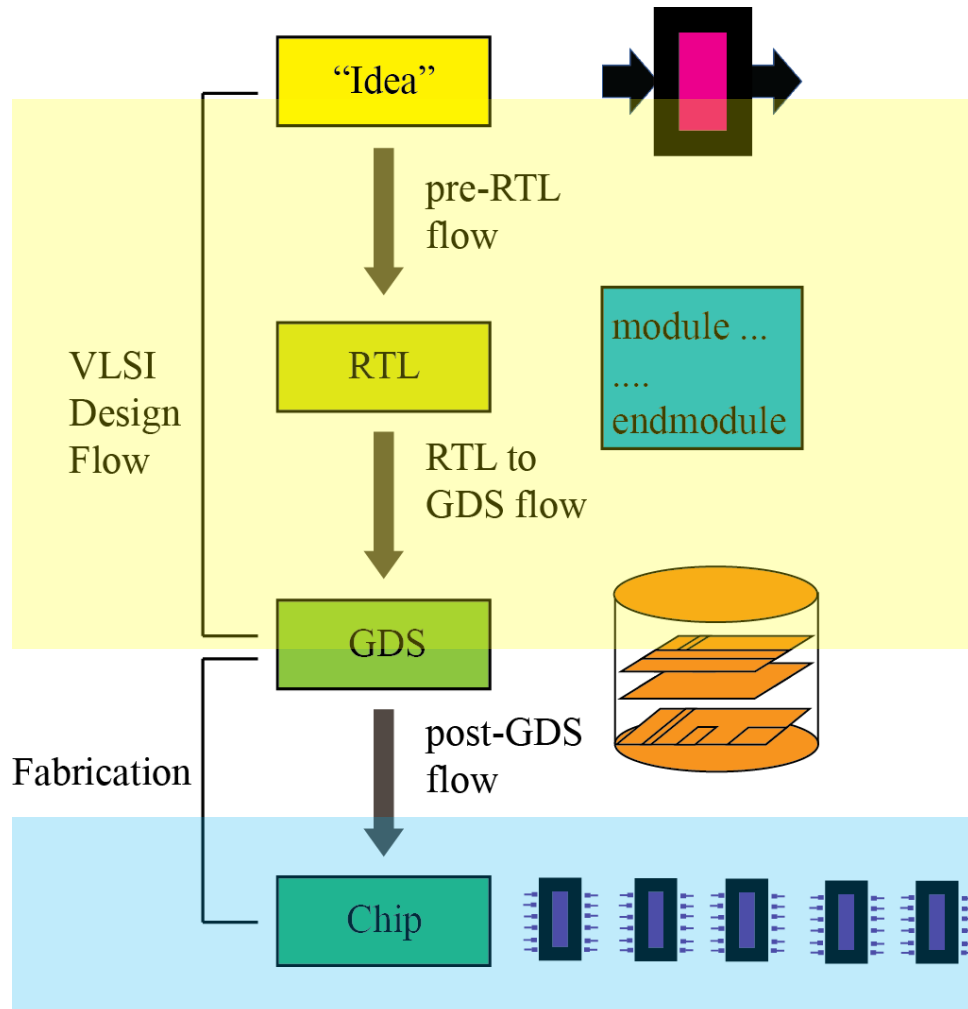
- ensure that the layout is functionally equivalent to the original netlist

Overview of RTL to GDS Flow



Testing

VLSI Design Flow: Testing



- **Verification** : Ensures that GDS represents the circuit correctly such that original specification is met.

- **Testing** : Ensures that fabricated chip does not have any manufacturing defect

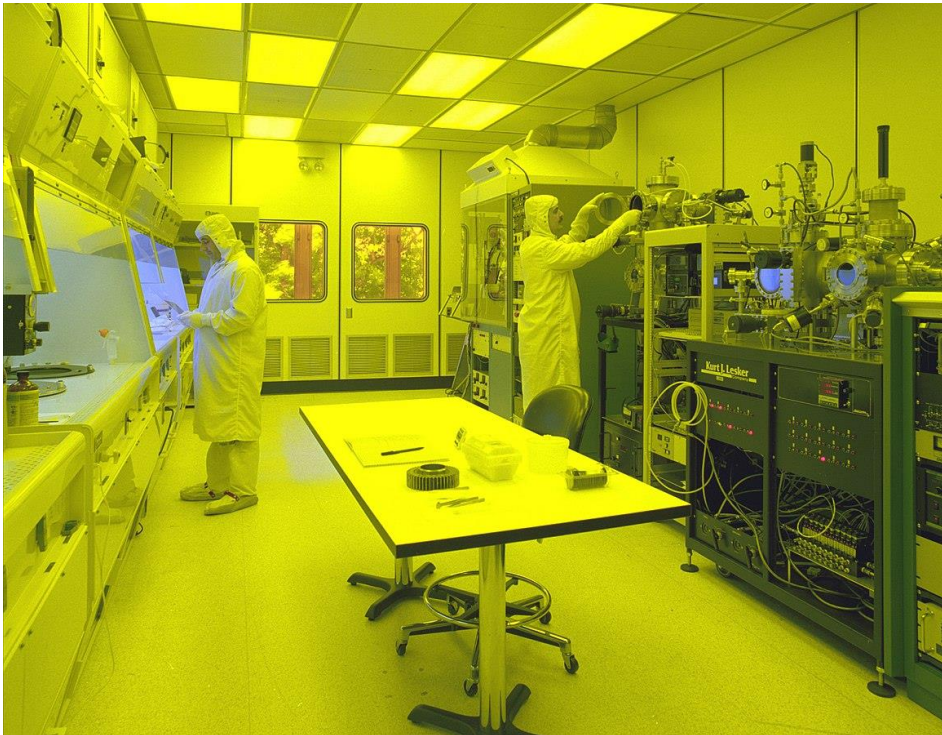
Why testing is important during designing?

- RTL to GDS flow ensures that the chip that is manufactured can be easily tested (**Testability**)
- RTL to GDS flow ensures that if some defect is found then the problem can be easily **diagnosed (debugged)** and fixed

Manufacturing Defects: Origin

Defects

- Physical imperfections in a fabricated chip that are of permanent nature



Source:

https://commons.wikimedia.org/wiki/File:Clean_room.jpg

Uploaded by Duk 08:45, 16 Feb 2005 (UTC), Public domain, via Wikimedia Commons

Origin of defects

- Statistical deviations in material properties
- Finite tolerances in process parameters
- Airborne particles, and undesired chemicals
- Deviations in mask features

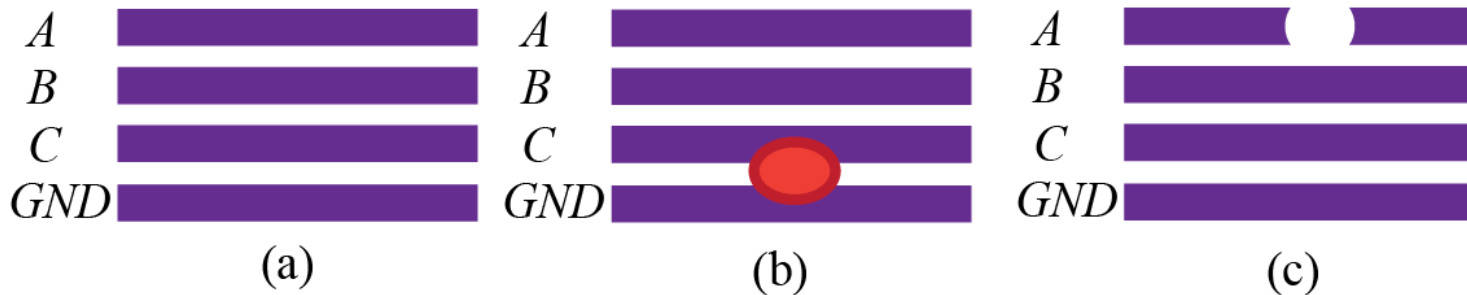
Large area and Spot defects

- Large area defects: simpler to eliminate
- Small area defects, of random nature
 - Inevitably appears in chip fabrication
 - Increases with increase in die area
 - Primary concern for testing

Manufacturing Defects: Manifestation

Manifestation of defects

- Short-circuit and open-circuit (functional failure)
- Change in circuit parameters (such as delay)



Faults:

- Testing focuses on finding defects that can be problematic for a circuit behavior
- We model defect (physical phenomenon) using **faults** (circuit model)

Distortions

- Photolithography can produce distorted features on a die
- These are inevitable due to optical effects, etc.
- Needs to be handled using suitable techniques
- Testing does not detect these distortions

Inconsequential flaws

- Deviations of a fabricated circuit from the ideal, but not causing any measurable change
- E.g.: if the size of particle is too small
- **Testing does not detect these inconsequential flaws**

Quality of Process: Yield

Yield

- Fraction of die on a wafer that are *good* or without any fatal manufacturing defect
- Usually expressed in percentage
- E.g.: If 300 dies are good out of 400 dies manufactured, then yield is:

$$= \frac{300}{400} \times 100\% = 75\%$$

Yield and process technology

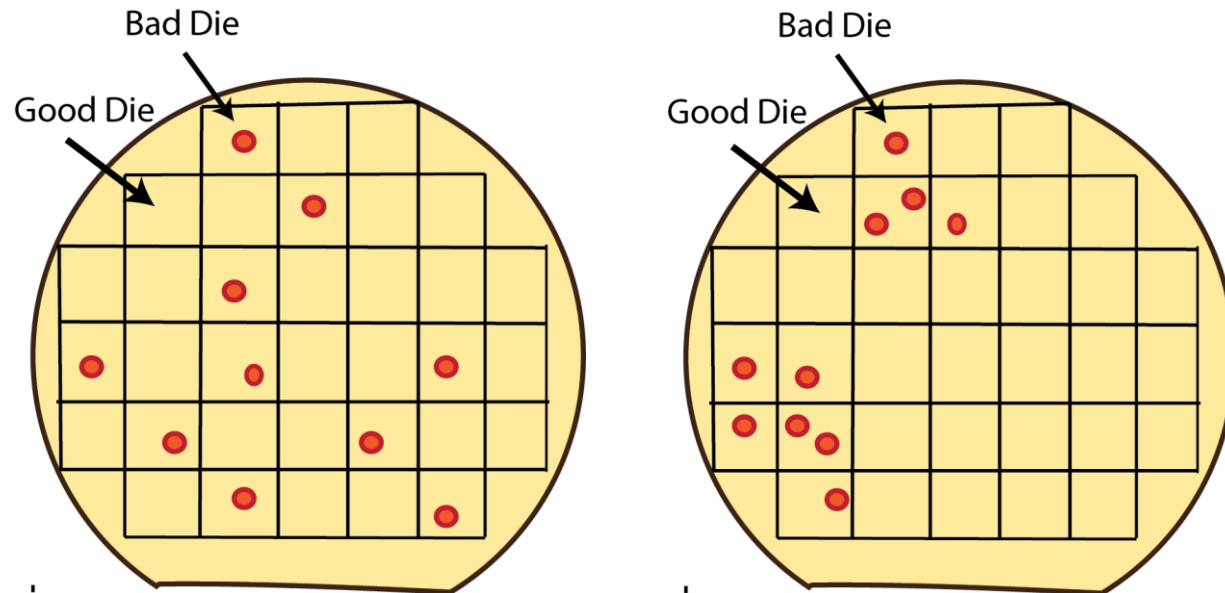
- Function of complexity and maturity of the process
- Desirable > 50%, and often > 90%

Factors affecting yield

1. **Die area:** when area increases, yield falls
2. **Defect density:** average number of defects per unit of chip area (depends on process)
3. **Clustering:** distribution of defects on the chip area

Yield: Dependency on clustering

Which is better for yield, when defects are clustered (defects lying in small region) or unclustered (same number of defects distributed over a larger region)?



Let us compute the yield in two cases shown.

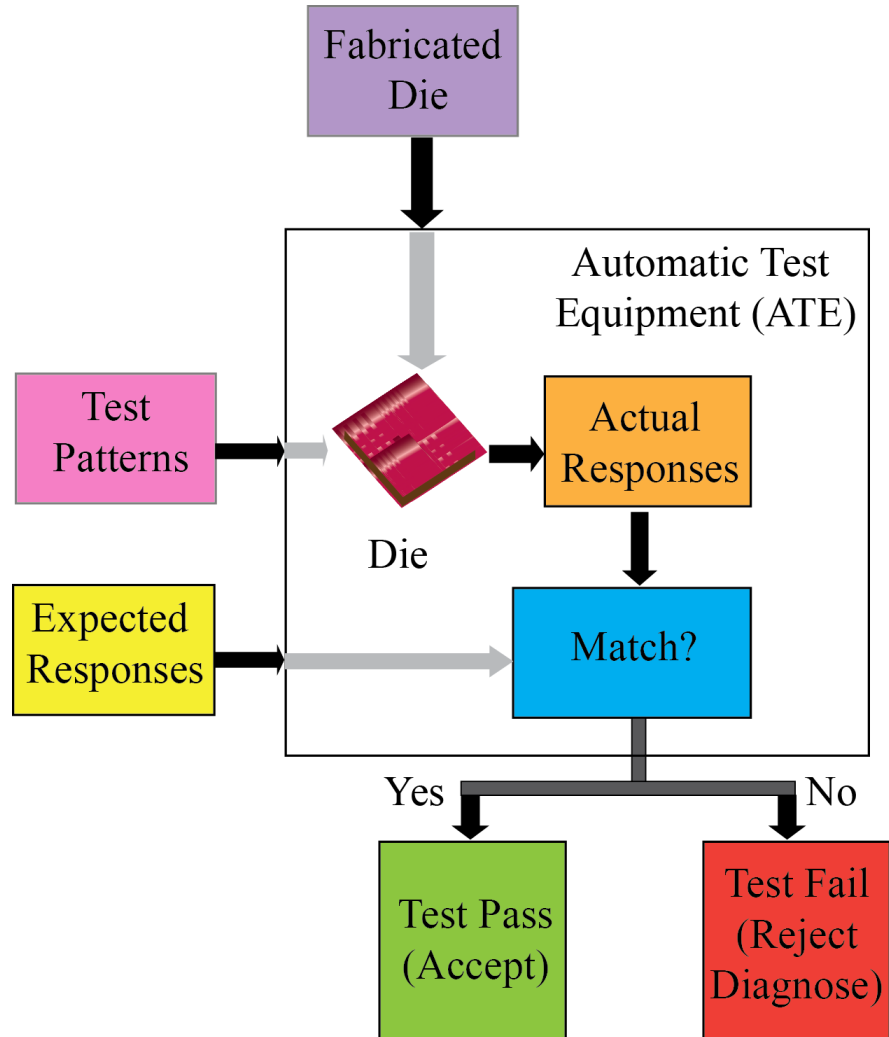
Unclustered defects

$$\text{Yield} = \frac{24}{34} \times 100 = 71\%$$

Clustered defects

$$\text{Yield} = \frac{26}{34} \times 100 = 76\%$$

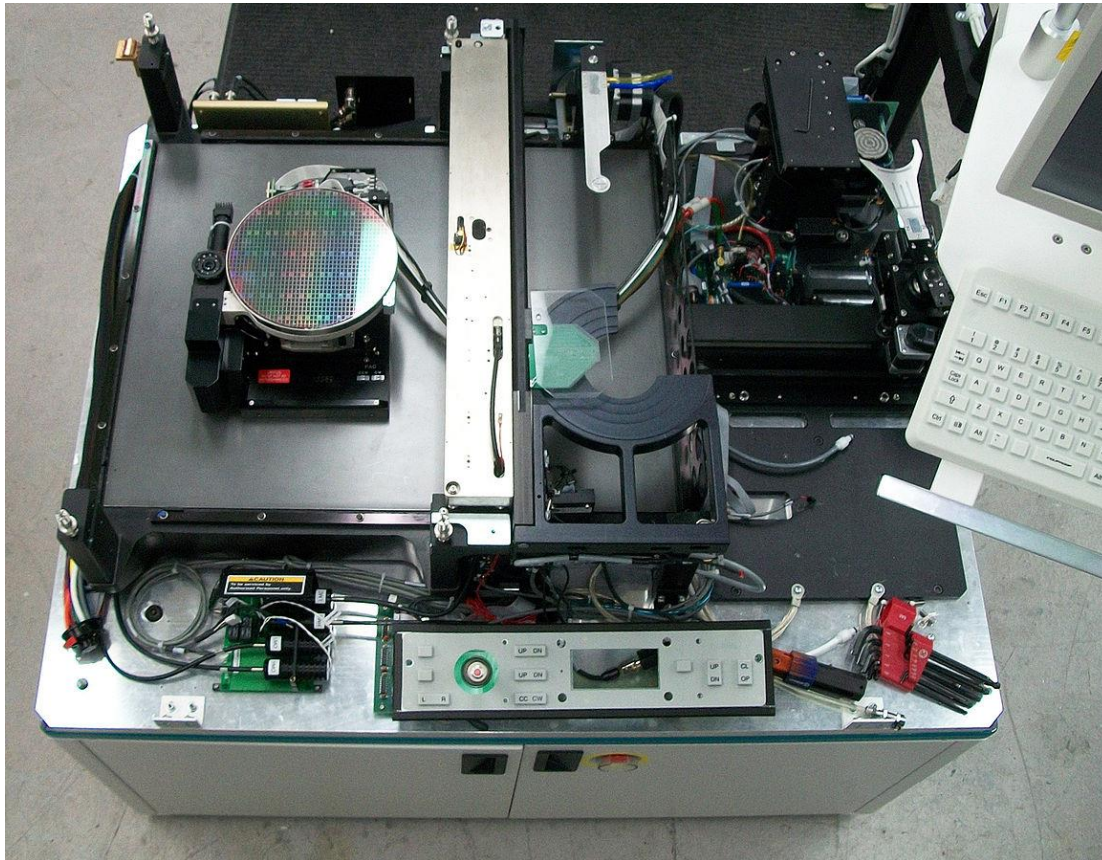
Testing Technique



- Test Patterns are applied to the manufactured chip
- Actual responses are compared with the expected response
- If comparison FAILs $\Rightarrow$ The chip has some defect(s)

- The failed chip can be diagnosed to find the root cause of the problem
- After diagnosis corrective measures can be taken to reduce the defects

Automatic Test Equipment (ATE)



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Automatic Test Equipment (ATE)

- Most important component of testing
- Consists of: test head, probe cards with probe needles
- Probe needles make contact with the test pads on the design under test (DUT)
- Test program controls all operation

Fault Coverage and Defect Level

We measure the quality of testing using a parameter called **fault coverage**

- **Fault coverage:** measures the ability of the set of test patterns to detect a class of faults
- $$\text{Fault coverage} = \frac{\# \text{ faults detectable}}{\# \text{ faults possible}}$$
- Fault coverage is a measure of *quality of testing*

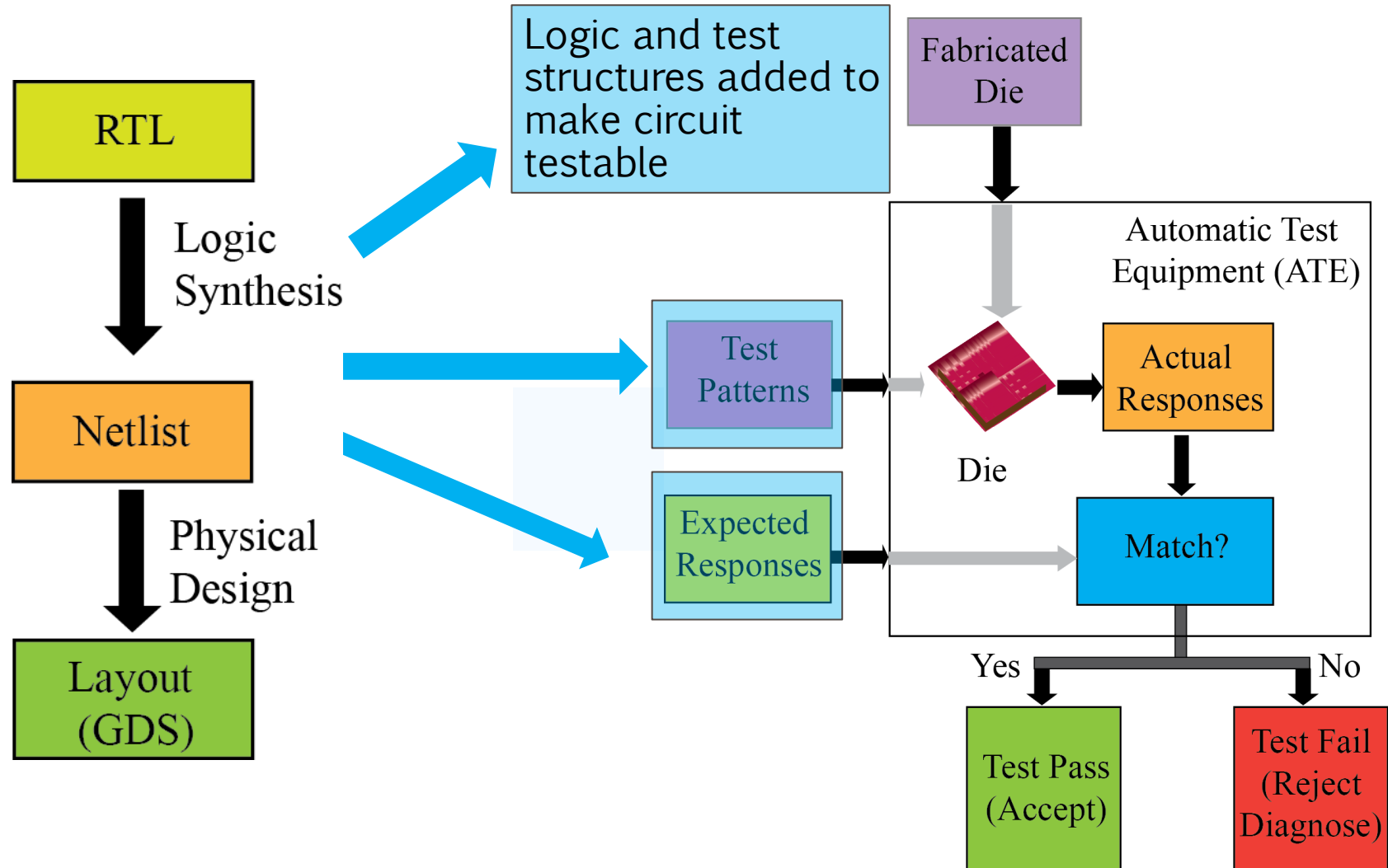
- Practically achieving 100% fault coverage is challenging
- We try to attain more than 99% coverage
- Some faulty products can reach the end user because the employed set of test patterns may not cover/test all possible faults

- Perceived quality of a chip strongly depends on: **fault coverage** and **yield**
- Quality of chip is measured by a quantity called *defect level*

Defect Level (DL)

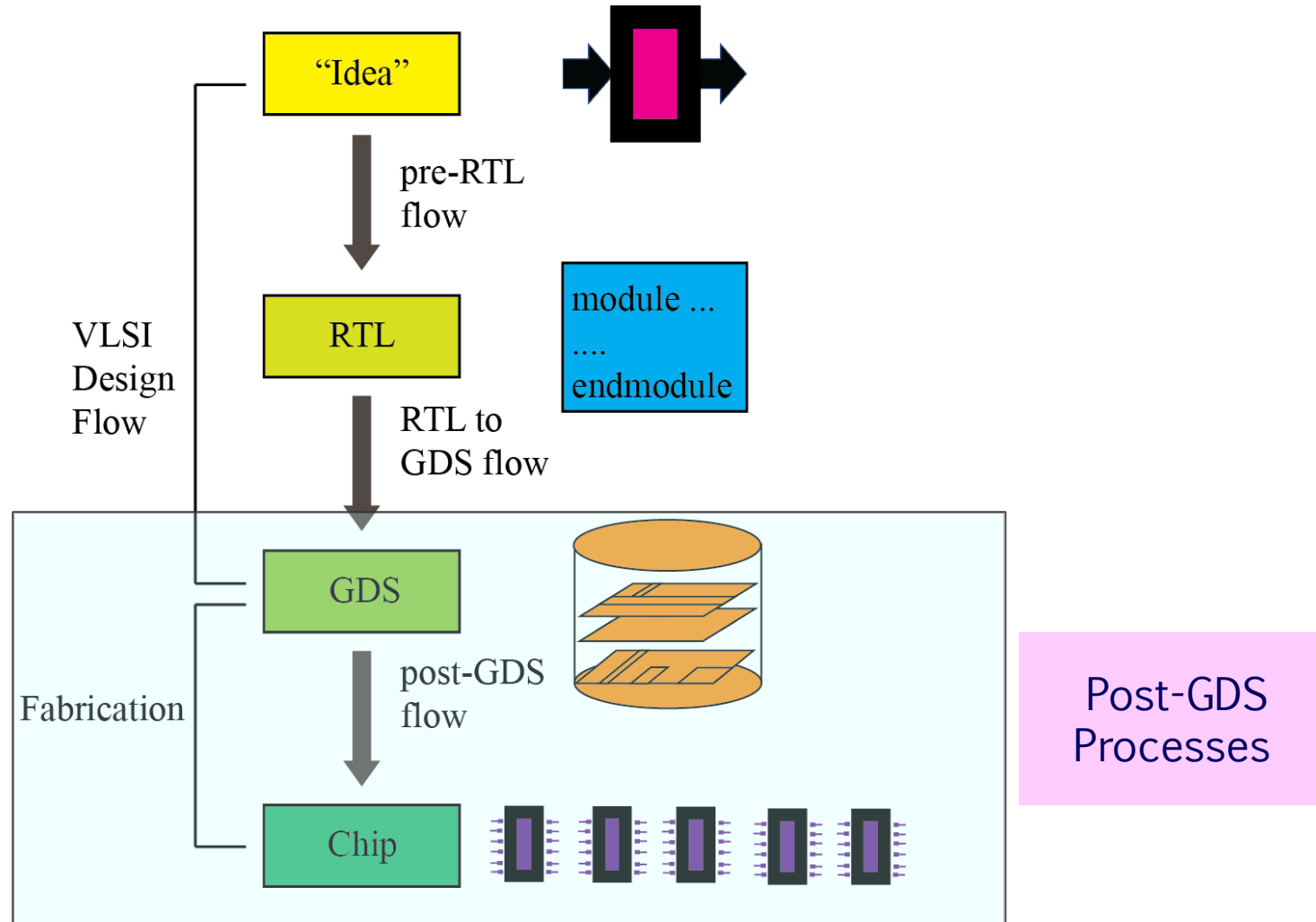
- Ratio of chips that are “bad” or faulty among the chips that have PASSED the tests
 - Measured in parts per million (ppm)
-
- DL is a measure of effectiveness of the tests (if test is fully effective, then $DL = 0$)
 - For commercial chips $DL < 500 \text{ ppm}$

Design For Test (DFT)



Overview of Post-GDS Processes

Post-GDS Processes



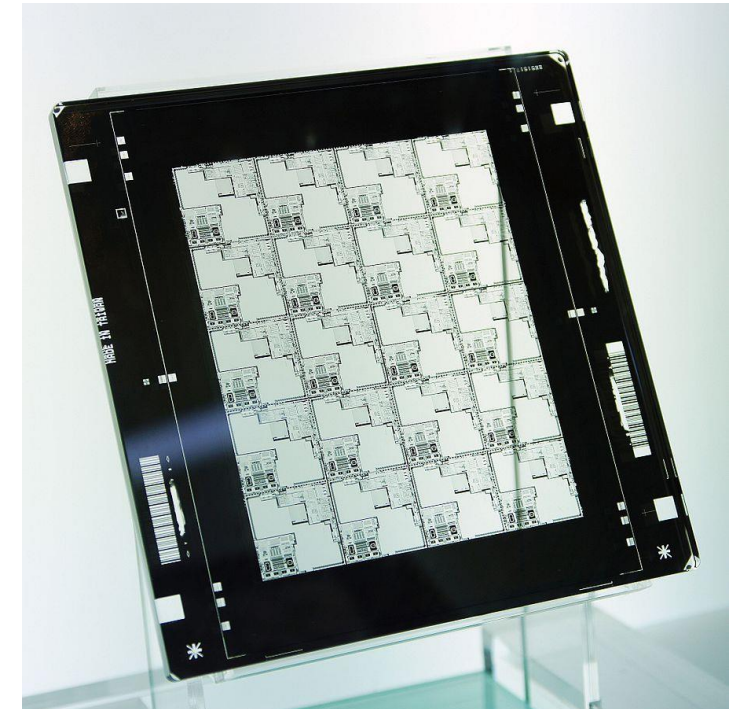
Mask Fabrication

- Mask is the replica of the patterns on a given layer of the layout created on a substrate (glass) for transferring pattern during photolithography

- We need to fabricate mask before fabricating the corresponding IC

Mask fabrication typically involves following steps:

- Data preparation
- Mask writing and chemical processing
- Quality checks and adding protections



Source:

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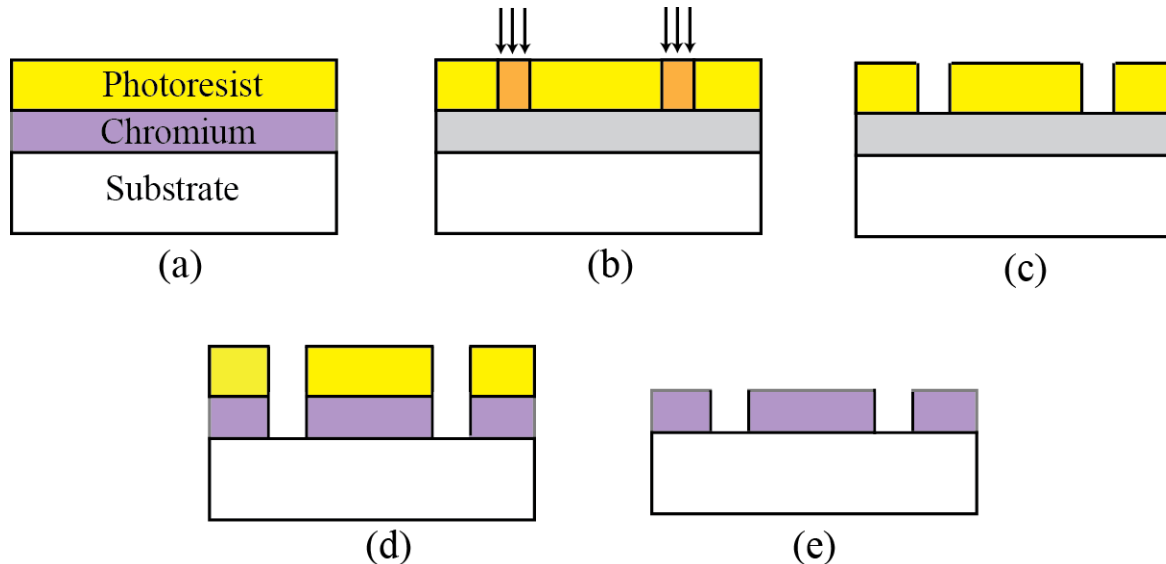
Mask Writing, Quality Checks and Protection

Data preparation:

- Translate the layout-specified mask information to a format comprehended by a mask writing tool
 - convert complicated polygons to simpler rectangles and trapeziums (fracturing)
- Augment mask data to enhance resolution

Mask Writing

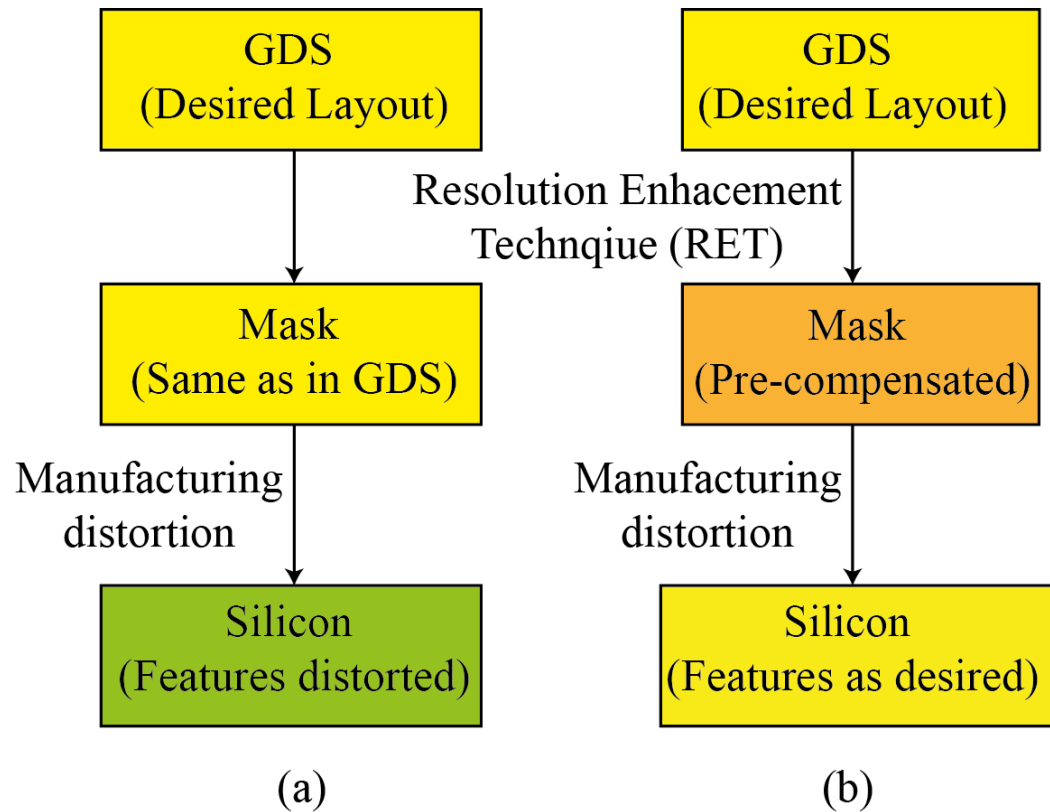
- Start with chromium and photoresist coated on glass or quartz (**blank**)
- Pattern written on the **blank** by exposing to LASER or electron beam
- Photoresist is developed, chromium is etched and then photoresist is stripped



Quality checks and protection

- Inspect for defects by scanning its surface and comparing it with the reference image
- If a defect is beyond tolerance, we repair it with the help of LASER
- Finally, protective cover called **pellicle** is applied

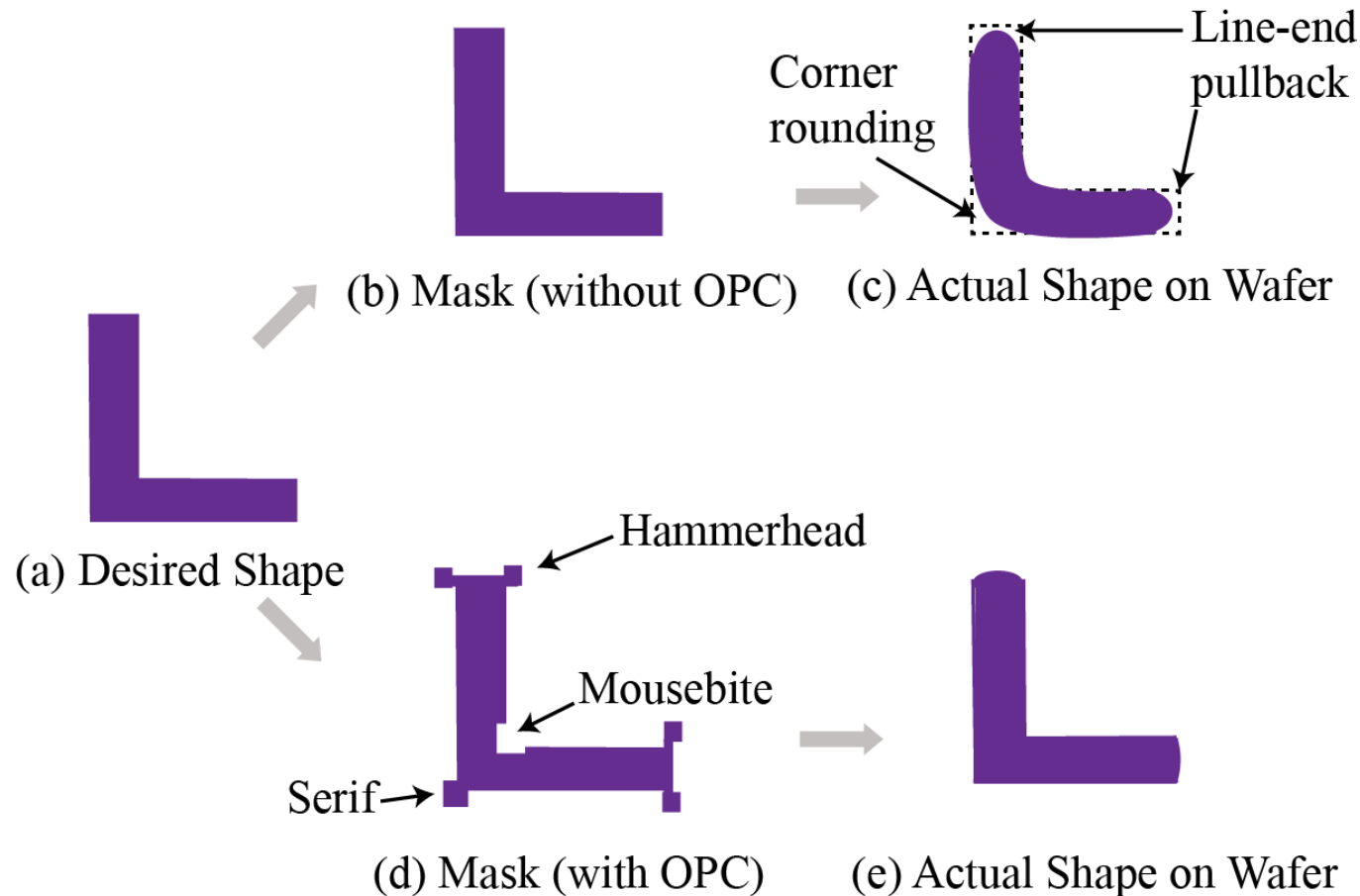
Resolution Enhancement Techniques (RET)



- If mask is patterned exactly as in GDS, then features obtained on silicon is distorted
 - Due to physical effects such as diffraction

- Resolution Enhancement Techniques (RET): mask is pre-compensated such that the features obtained on the mask is same as is desired
 - Examples: Optical Proximity Correction (OPC) and Double/Multi-Patterning

RET: Optical Proximity Correction (OPC)



- Printing mask patterns smaller than the light wavelength produces distortions
- E.g.: rounding of corners and line-end pullback.

Optical Proximity Correction (OPC)

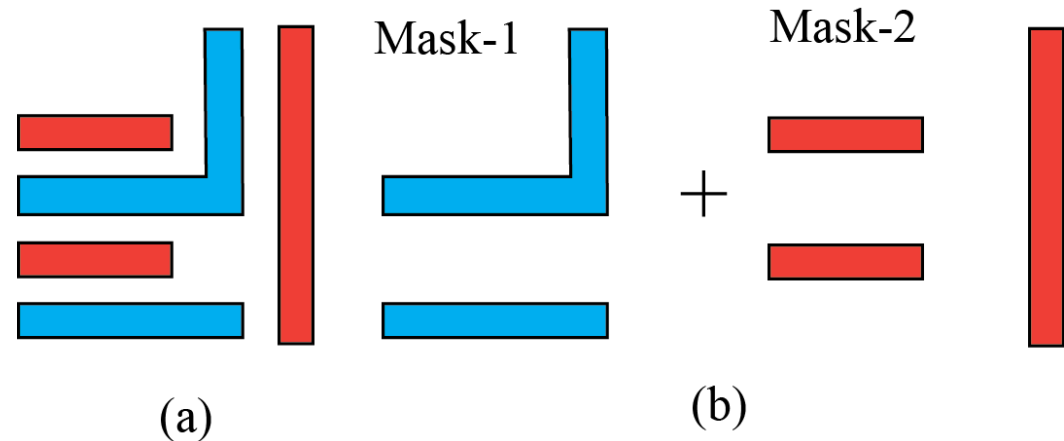
- Add appropriate serifs, hammerheads, and mouse bites to the mask
- Improves the resolution of photolithography by compensating errors due to diffraction etc.

RET: Double- or Multiple-patterning

- Limited resolution of photolithography, pose problems in printing closely spaced features on a die (leads to overlaps)
- We can solve the problem by increasing the spacing between features printed at a time

Double- or Multi-patterning

- Decompose a closely spaced layout into two or more layouts (assign colors to features)
- We use different masks and different exposures for layout features of different colors.
- Each exposure needs a lower resolution due to decreased feature or pattern density.



References

- S. Saurabh, “Introduction to VLSI Design Flow”. Cambridge: *Cambridge University Press*, 2023.